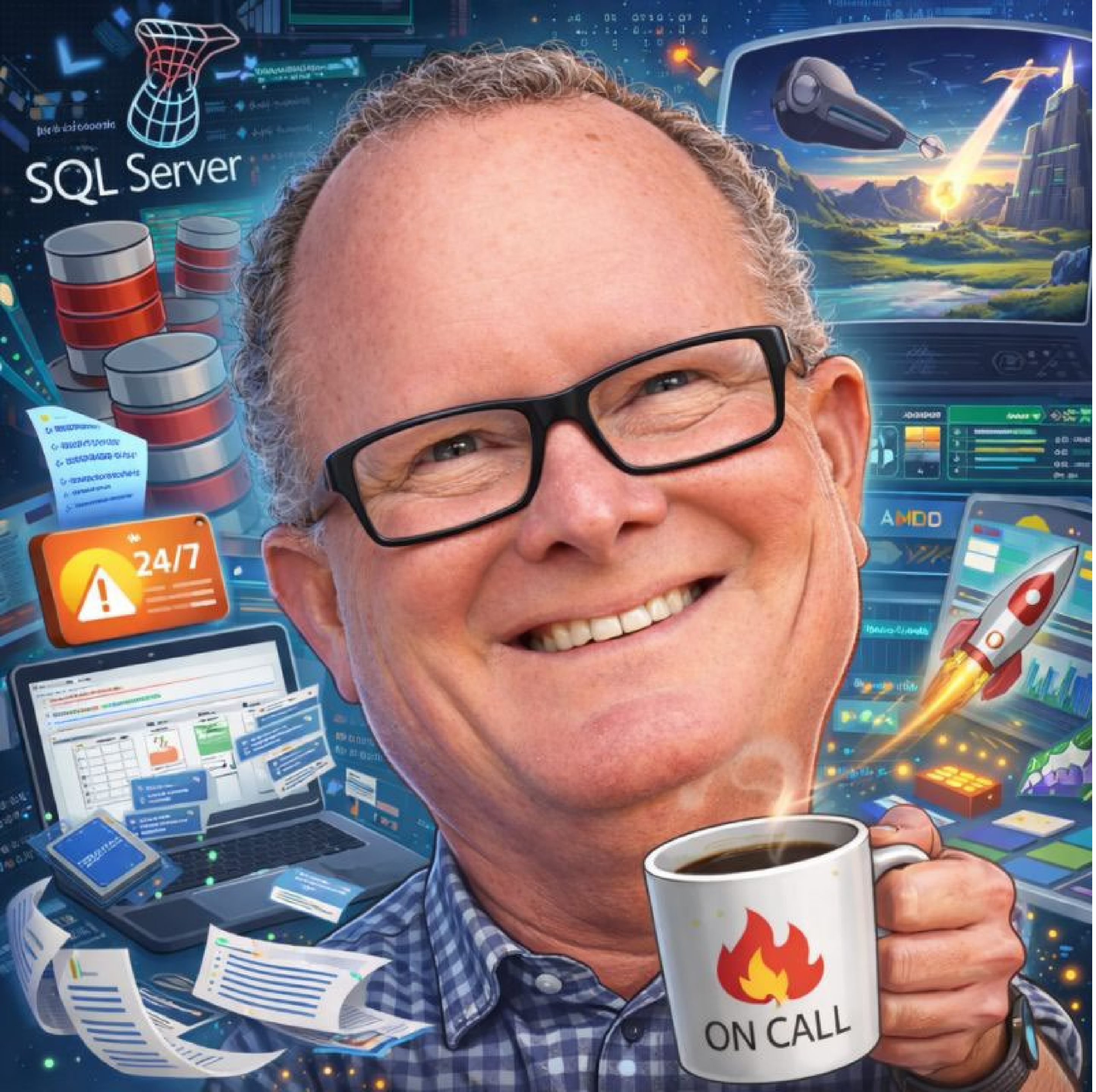


Pandas Without Limits

Accelerating Pandas Using RAPIDS

todd.kleinhans@gmail.com
https://www.linkedin.com/in/todd-kleinhans-981920/

Outline

- What are Pandas?
- What is NVIDIA RAPIDS and are there some limitations?
- Compare and contrast Pandas to RAPIDS cuDF
- Ways to run RAPIDS cuDF from free to $$$
- Demos using Jupyter Notebooks for both cloud and on-premises
- Resources
- Q & A

What are Pandas?

The name pandas comes from the term panel data referring to tabular data like in a spreadsheet or a table in a relational database and implemented in Python

Restaurant location	
Diner	(4,2)
Diner	(4,3)
Diner	(5,4)
Diner	(5,5)

Yelp Score	
02/18	90
05/18	100
04/18	55
01/18	76

RAPIDS

GPU Accelerated Data Science

QUICK START

NEW: CUML NOW ACCELERATES SCIKIT-LEARN | NEW: POLARS GPU ENGINE | CUDF NOW PRE-INSTALLED IN GOOGLE COLAB | VECTOR SEARCH NOW WITH CUVS | RAPIDS 26.04 RELEASED

WHAT IS RAPIDS

RAPIDS provides unmatched speed with familiar APIs that match the most popular PyData libraries. Built on state-of-the-art foundations like **NVIDIA CUDA** and **Apache Arrow**, it unlocks the speed of GPUs with code you already know. **Jump to About Section ↗**

WHY USE RAPIDS

RAPIDS allows fluid, creative interaction with data for everyone from BI users to AI researchers on the cutting edge. GPU acceleration means less time and less cost moving data and training models. **Jump to RAPIDS Use Cases ↗**

OPEN SOURCE ECOSYSTEM

RAPIDS is Open Source and available on **GitHub**. Our mission is to empower and advance the open-source GPU data science data engineering ecosystem. **Jump to RAPIDS GitHub ↗**

PANDAS ACCELERATOR MODE FOR CUDF

Use cuDF pandas Accelerator Mode to speed up pandas workflows with zero code change. **Learn More on the Accelerator Mode Page ↗**

POLARS GPU ENGINE POWERED BY CUDF

Accelerate Polars by enabling the GPU engine with zero code change. **Learn More on the Launch Page ↗**

ACCELERATED SCIKIT-LEARN WITH CUML

Run machine learning models faster with zero code change. **Learn More on the Accelerated ML Page ↗**

NETWORKX SUPERCHARGED BY CUGRAPH

Speed up your large-scale graph workflows with zero code change. **Learn More on the nx-cugraph Page ↗**

HOW IT WORKS

UNDER THE HOOD

When `cudf.pandas` is enabled, `import pandas` (or any of its submodules) imports a magic module, rather than "regular" pandas. This magic module contains proxy types and proxy functions:

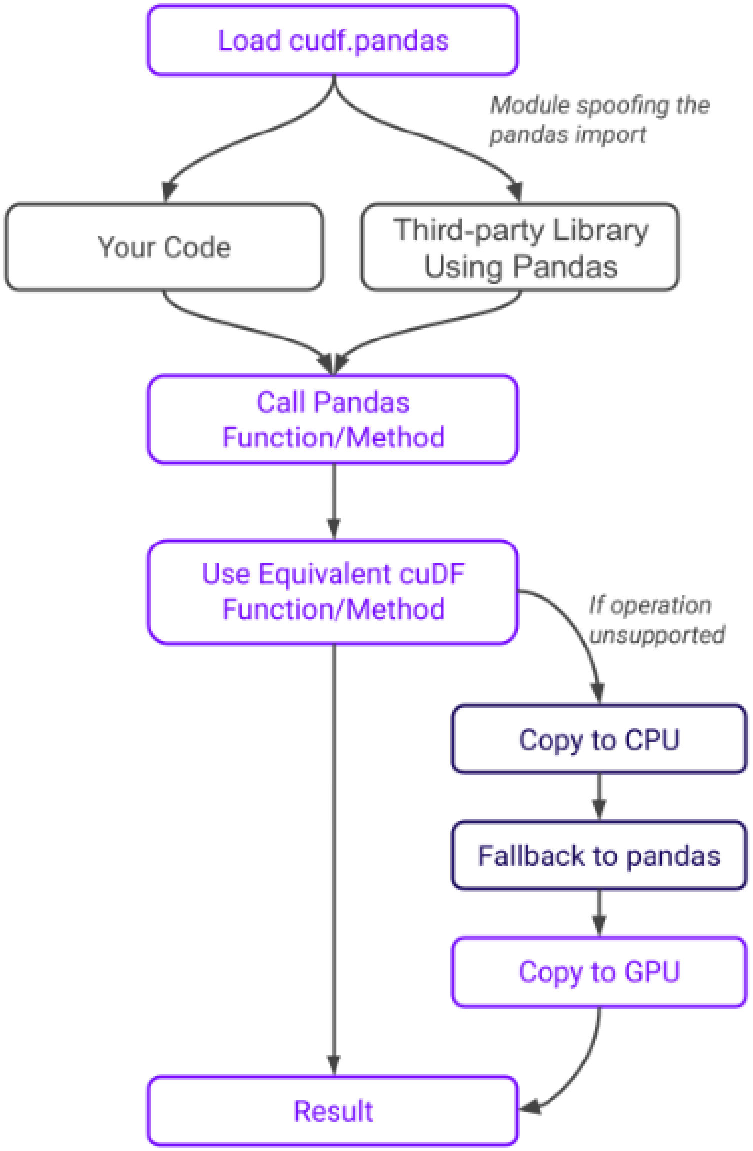
```
In [1]: %load_ext cudf.pandas
In [2]: import pandas as pd

In [3]: pd
Out[3]: <module 'pandas' (ModuleAccelerator(fast=cudf, slow=pandas))>
```

Operations on proxy types and functions execute on the GPU where possible and on the CPU otherwise, synchronizing under the hood as needed. This applies to pandas operations in both your code and in third-party libraries you may be using.

All `cudf.pandas` objects are a proxy to either a GPU (cuDF) or CPU (pandas) object at any given time. Operations are first attempted on the GPU (copying from CPU if necessary). If that fails, the operation is attempted on the CPU (copying from GPU if necessary).

When using `cudf.pandas`, cuDF's pandas compatibility mode is automatically enabled, ensuring consistency with pandas-specific semantics like default sort ordering.

EXECUTION FLOW



And Now, The Sad Trombone…

- RAPIDS will ONLY run on NVIDIA RTX+ hardware (sorry AMD)
- RAPIDS will ONLY run on Linux
- “How are you able to run it on a Windows 11 laptop? ;)”
  - HINT: Windows Subsystem for Linux (WSL2) & Docker containers

Compare & contrast Pandas to RAPIDS cuDF

Area	**Traditional pandas**	**RAPIDS cuDF**
Compute device	CPU	NVIDIA GPU
Parallelism	Limited CPU cores	Thousands of CUDA cores
Backend	NumPy + C extensions	CUDA + GPU kernels
Best dataset size	MBs to low GBs	GBs to TB-scale pipelines
Memory location	System RAM	GPU VRAM
Execution style	Mostly serial/vectorized CPU	Massive parallel vectorized GPU
Strength	Ease of use, ecosystem	Extreme throughput
Weakness	CPU bottlenecks	GPU memory constraints
Ideal workloads	Small/medium analytics	ETL, joins, aggregations, ML prep
Compatibility	Native pandas	pandas-like API

Where RAPIDS Wins BIG

RAPIDS shines when you have dataframes which use:

- Large joins
- Grouping and sorting
- ETL pipelines
- String processing
- Feature engineering
- Time-series aggregation
- Repeated workloads
- Need to scale

Where RAPIDS Is NOT Ideal

- Small datasets
- Unsupported APIs (not 100% - has most commonly used functions)
- GPU Memory constraints

Ways to run RAPIDS cuDF From Free to $$$

- Google Colab (free)
  - Colab Pro ($9.99/mo) & Pro+ ($49.99/mo) & Enterprise (usage)
- Amazon SageMaker Studio Lab (free)
  - Can upgrade to SageMaker Studio (not Lab)
- Paperspace (now Digital Ocean) (free)
  - Pro ($8/mo) & Growth ($39/mo) & Enterprise (usage)
- NVIDIA Launchpad
  - Free for short term use & WIDE variety of models and options
- Cloud Vendors (Azure, AWS, GCP, IBM, Oracle)

Ways to run RAPIDS cuDF From Free to $$$ (Local)

- Your RTX laptop or desktop w/ RTX graphics card
  - Windows Subsystem for Linux (WSL2) & Docker containers
- Upgrade to a newer RTX graphics card
  - Old RTX 3090 $2200 (10496 cores) vs today RTX 4090 $1700 (16384 cores)
  - NVIDIA RTX Pro 6000 PCIe 96GB VRAM $9000
- ASUS Ascent GX10 128GB VRAM $3500 (200B parameters)
  - Can network two together for 405B parameters
- NVIDIA DGX Spark 128GB VRAM $4500

- Enterprise Grade (server-based)
- NVIDIA H200 GPU $31,000+

Demos – Colab and on-premises

Resources

How to Spot (and Fix) 5 Common Performance Bottlenecks in pandas Workflows 22Aug25
https://developer.nvidia.com/blog/how-to-spot-and-fix-5-common-performance-bottlenecks-in-pandas-workflows/

7 Drop-In Replacements to Instantly Speed Up Your Python Data Science Workflows 01Aug25
https://developer.nvidia.com/blog/7-drop-in-replacements-to-instantly-speed-up-your-python-data-science-workflows/

3 pandas Workflows That Slowed to a Crawl on Large Datasets—Until We Turned on GPUs 18Jul25
https://developer.nvidia.com/blog/3-pandas-workflows-that-slowed-to-a-crawl-on-large-datasets-until-we-turned-on-gpus/

Mastering the cudf.pandas Profiler for GPU Acceleration 30Jan25
https://developer.nvidia.com/blog/mastering-the-cudf-pandas-profiler-for-gpu-acceleration/

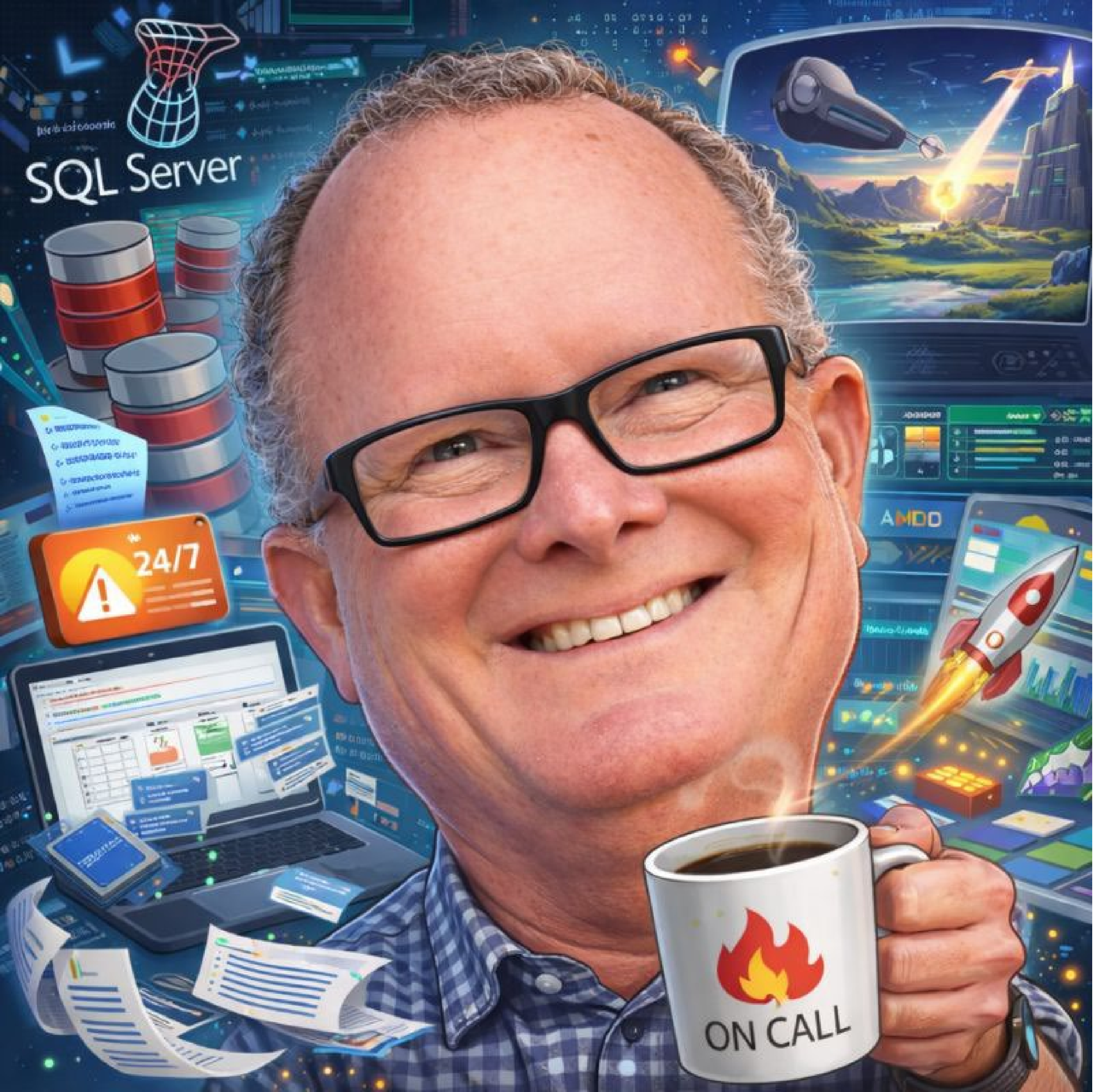
10 Minutes to RAPIDS cuDF's pandas accelerator mode (cudf.pandas) LOGIN TO GMAIL
https://colab.research.google.com/github/rapidsai-community/showcase/blob/main/getting_started_tutorials/cudf_pandas_colab_demo.ipynb

RAPIDS cuDF Instantly Accelerates pandas up to 50x on Google Colab
https://developer.nvidia.com/blog/rapids-cudf-instantly-accelerates-pandas-up-to-50x-on-google-colab/?ncid=em-news-294101

Getting Started with Google Colab
https://colab.research.google.com/github/googlecolab/colabtools/blob/main/notebooks/Getting_started_with_google_colab_ai.ipynb

Making the Most of your Colab Subscription (Note availability from within Visual Studio Code!)
https://colab.research.google.com/notebooks/pro.ipynb

Accelerate Data Science Workflows with Zero Code Changes
https://learn.nvidia.com/courses/course-detail?course_id=course-v1:DLI+T-DS-03+V1



Q & A

Accelerating Pandas Using RAPIDS

todd.kleinhans@gmail.com
https://www.linkedin.com/in/todd-kleinhans-981920/